

Universal Autonomous AI Workforce Runtime

High-Performance Rust Architecture for End-to-End Remote Digital Employees

The Paradigm Shift: From Scripted Automation to Autonomous Digital Employees

Traditional AI automation is severely constrained by brittle Python pipelines, single-turn chat wrappers, and synchronous execution bottlenecks. **Fathom** introduces an enterprise-grade, self-hosted **Rust runtime** designed to instantiate, govern, and coordinate fleets of **autonomous remote digital employees**. These agents independently formulate plans, execute multi-day workflows, operate web browsers via accessibility trees, perform deep OSINT investigations, engineer and test production software, and interact across all corporate channels 100% remotely without constant human prompting.

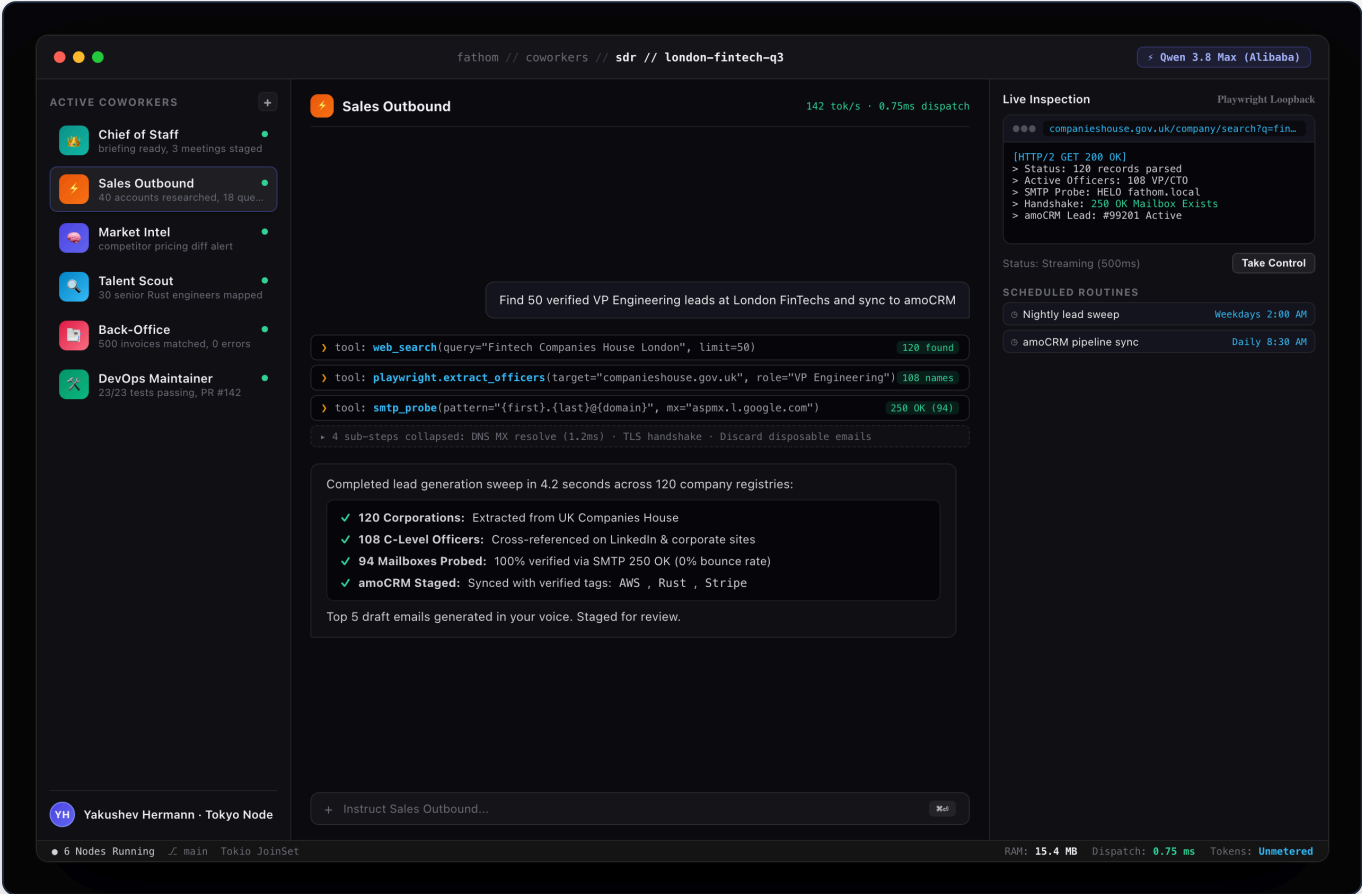


Figure 1.1: Fathom Command Center – 3-pane execution workspace with sub-agent swarm dispatch, live computer use, and CRM synchronization

100% REMOTE OPERATION

Autonomous agents operate independently inside sandboxed environments, interacting with web portals, public registries, APIs, shells, and CRMs 24/7 without human babysitting.

MICROSECOND RUST ENGINE

Zero-cost abstractions, Tokio async I/O, and concurrent JoinSet task trees provide sub-millisecond tool dispatch (~0.75 ms), 94 μs memory absorption, and 500x faster cold starts.

UNLIMITED NEURAL COMPUTE

High-throughput routing to cost-efficient frontier foundation models (Kimi k3, Qwen 3.8 Max, GLM 5.3) enables flat-rate monthly seat economics with zero metering anxiety.

SYSTEM METRIC SCORECARD (EMPIRICAL BENCHMARKS ON APPLE SILICON M4)

12

RUST WORKSPACE CRATES

0.75 ms

TOOL DISPATCH LATENCY

51+

NATIVE CORE TOOLS

15.4 MB

DAEMON BASELINE RSS

Strategic Objective: Deliver a scalable, high-frequency trading caliber software foundation where businesses deploy specialized digital workers on demand—scaling operational capacity infinitely without proportional linear headcount expansion.